

Study

Simplify

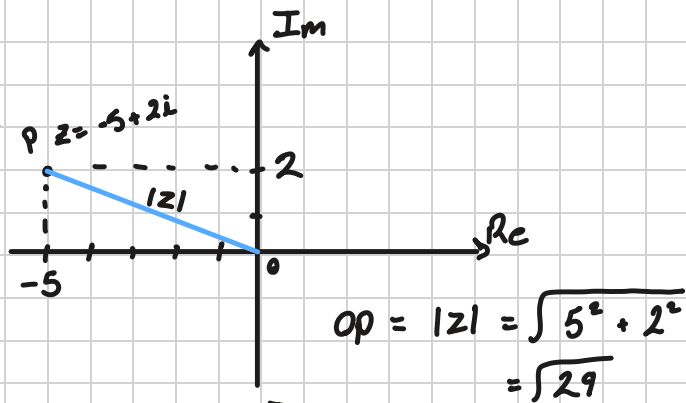
$$a) \frac{2+3i}{4-i} \times \frac{4+i}{4+i} = \frac{(2+3i)(4+i)}{(4+i)(4-i)} = \frac{8-3+(2+12)i}{16+1} = \frac{5}{17} + \frac{14}{17}i$$

$$b) \frac{i}{1+i\sqrt{3}} = \frac{i(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{\sqrt{3}+i}{1+3} = \frac{\sqrt{3}}{4} + \frac{1}{4}i$$

$$\textcircled{1} \quad z = -5 + 2i$$

$$\operatorname{Re}(z) = -5$$

$$\operatorname{Im}(z) = 2$$



$$\textcircled{2} \quad z = 4 - i$$

$$\operatorname{Re}(z) = 4$$

$$\operatorname{Im}(z) = -1$$

